

Logistic Regression Analysis

=== GAME 1 ANALYSIS ===

Churn Distribution:

Not Churned: 1928 (7.4%)

Churned: 24028 (92.6%)

Original distribution: [1542 19222]

Balanced distribution: [3402 15197]

New churn rate: 81.7%

Optimal threshold: 0.019 (F1: 0.965)

=== Model Evaluation ===

AUC Score: 0.7884

Optimal Threshold: 0.019

Classification Report:

	precision	recall	f1-score	support
0	0.71	0.19	0.30	386
1	0.94	0.99	0.97	4806
accuracy			0.93	5192
macro avg	0.83	0.59	0.63	5192
weighted avg	0.92	0.93	0.92	5192

=== GAME 2 ANALYSIS ===

Churn Distribution:

Not Churned: 2197 (22.2%)

Churned: 7684 (77.8%)

Original distribution: [1757 6147]

Balanced distribution: [339 3968]

New churn rate: 92.1%

Optimal threshold: 0.418 (F1: 0.891)

=== Model Evaluation ===

AUC Score: 0.7451

Optimal Threshold: 0.418

Classification Report:

	precision	recall	f1-score	support
0	0.73	0.27	0.39	440
1	0.82	0.97	0.89	1537
accuracy			0.81	1977
macro avg	0.77	0.62	0.64	1977
weighted avg	0.80	0.81	0.78	1977

Game 1 Rankings (10 features):

1. activeDuration	: 1.4893
2. worstScore	: 1.2908
3. meanScore	: 1.2203
4. consecutivePlayRatio	: 0.9342
5. playCount	: 0.6744
6. sdScore	: 0.4266
7. bestScore	: 0.3720
8. bestSubMeanCount	: 0.2719
9. bestSubMeanRatio	: 0.1172
10. bestScoreIndex	: 0.1069

Game 2 Rankings (12 features):

1. highestPrice	: 3.6541
2. activeDuration	: 1.2314
3. sdScore	: 0.5009
4. bestScore	: 0.4626
5. bestSubMeanRatio	: 0.3920
6. worstScore	: 0.3129
7. playCount	: 0.2678
8. consecutivePlayRatio	: 0.2241
9. purchaseCount	: 0.1794
10. bestScoreIndex	: 0.0423
11. meanScore	: 0.0379
12. bestSubMeanCount	: 0.0147